

Boundaries and 'Not Sure'

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Save New Examples for Testing

Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Learning from Examples

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

If/Then Decisions

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Keep a stable prediction within its limits for Uncertainty and review. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Uncertainty and review. Proposed explanation: “Two possible values imply equal probabilities”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Transfer the method to distance

A paper model predicts far at a distance of at least 4 units, otherwise near, for whole-number distances 1 through 8. For each complete possibility set, compare predictions: 2 or 3; 3 or 4; 4 or 5. Identify the set requiring review before a single category can be stated. Design a relevant check and an unresolved ending. Limit the claims for the other two sets.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A sorter receives a clear purple card, a red card hidden by a sleeve and a card labelled both red and blue by two reviewers. Give a different reason for uncertainty in each case and a suitable next action.

6. Solve the given case

A sorter handles clearly red and clearly blue cards. Purple and very dark cards arrive.

How should it handle each uncertain case?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A sorter knows pencils and books. You show a lunch box. Should it force a known label or ask for help?
State a useful fallback.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 16 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Add a not-sure branch, missing-information prompt, and human review destination to the sorter.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A sorter receives a clear purple card, a red card hidden by a sleeve and a card labelled both red and blue by two reviewers. Give a different reason for uncertainty in each case and a suitable next action.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Describe the uncertainty precisely

For a smudged height that could be 5 or 6 cm, state what is known, what is unknown and the two predictions under tall at least 6. Choose a check that could distinguish the heights. Explain why rerunning the threshold comparison cannot do that.

My reasoning and evidence

13. W2 Keep a stable prediction within its limits

Assume the only possible heights are 8 and 9 cm. Apply tall at least 6 to each. Write a justified result statement and two things it does not establish. Do not invent a confidence percentage.

My reasoning and evidence

14. W3 Choose useful review evidence

A record says 60 but omits the unit. A reviewer suggests reading it as 60 cm because the model needs cm. Explain the problem, name the evidence to request and state the outcome if the unit cannot be established.

My reasoning and evidence

15. W4 Transfer the method to distance

A paper model predicts far at a distance of at least 4 units, otherwise near, for whole-number distances 1 through 8. For each complete possibility set, compare predictions: 2 or 3; 3 or 4; 4 or 5. Identify the set requiring review before a single category can be stated. Design a relevant check and an unresolved ending. Limit the claims for the other two sets.

My reasoning and evidence

16. W5 Review the reviewer’s conclusion

A reviewer cannot read the original tower card but chooses 6 cm because two models both predict tall when given 6. Explain why this fails to resolve the input. Write a more defensible review record and a next step.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Boundaries and 'Not Sure'

A fictional cubby selector accepts a block if width is at most 5 cm.

An unclear measurement means Block A is either 4 or 5 cm; Block B is either 5 or 6 cm.

These complete possibility sets contain all widths under consideration. Other dimensions are known to fit.

Possibilities and decisions

	Width 1 cm	Decision 1	Width 2 cm	Decision 2
Block A	4	fits	5	fits
Block B	5	fits	6	fails

Read a cell using both its row and column labels.

Boundaries and 'Not Sure'

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Block A possibilities

4 and 5 both satisfy the including equality width limit.



2. Block B possibilities

5 fits; 6 does not.



3. Compare decisions

A has one supported decision; B's possible decisions disagree.



4. Review B

Ask a named reviewer to remeasure B using the specified width unit.

CHANGE ONE THING

Narrow B's possibilities to 6 or 7 cm.

Explain the changed result using the matching lesson.

Which original block needs a better width before accepting or rejecting it? What should the reviewer do?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Add explicit review paths to the sorter, including unresolved input, responsible person and evidence needed.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Add explicit review paths to the sorter, including unresolved input, responsible person and evidence needed.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional cubby selector accepts a block if width is at most 5 cm.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Ask a named reviewer to remeasure B using the specified width unit.

1. Block B possibilities
2. Compare decisions
3. Block A possibilities
4. Review B

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Review B Ask a named reviewer to remeasure B using the specified width unit. Uncertainty about a number need not always create uncertainty about the decision. Every allowed width for A fits, so its width uncertainty does not affect acceptance. B straddles the boundary. Record both possibilities and request a better measurement instead of selecting the one that makes a preferred outcome.

Practice 2: explain the deciding evidence

Changed-case practice: Narrow B's possibilities to 6 or 7 cm. Which conclusion is supported by the supplied facts?

1. Both fail the width limit; rejection no longer depends on resolving their difference.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Both fail the width limit; rejection no longer depends on resolving their difference. All remaining possibilities lead to the same decision.

Repair a missing building block

Review Save New Examples for Testing (ai-1/15). Goal: Separate examples used to choose a model from examples reserved for its final evaluation. Short practice: Does 4/4 on the replay replace the original 3/4 result? Explain.. Return to this week and explain what you changed.

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing predictions and counting disagreements for every candidate. Short practice: Which new measured length would reveal disagreement between the candidates?. Return to this week and explain what you changed.

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which input detects the difference between the two written conditions? Explain its entire run.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
